

12. Analysis C: true vs sensed area

Sensed area is compared with two references. One is the true area of the as-built feature. The other is the area that stereo geometry allows the sensor to see. The second matters for cutouts: part of the back plate seen through a hole is visible to only one camera, or is not lit by the projector, so no stereo sensor could read it.

1. Feature window. For each feature, frame, and pose, project the as-built outline into the image using the registered pose. The window is a circle of radius $D/2 + \text{BOUNDARY_BAND_HALF_WIDTH_PX}$ (in px at that Z) around the projected center.
2. Reference planes. Fit the front plane to an annulus of front surface outside the window. For a cutout this is the front plate. For a disk it is the registered disk plane, because small disk faces have too few pixels to fit. Fit the back plane to an annulus of back plate beyond the shadow band.
3. Pixel classes. Compute h as in 11.1. Inside the window, each pixel is a front read ($h \geq 0.5$), a back read ($h < 0.5$), or a no-read.
4. Sensed area per frame. For a disk, sensed pixels are the front reads connected to the projected center. For a cutout, they are the back reads connected to it. Each pixel's area on the feature plane is $a = Z_{\text{plane}}(u, v)^2 / (f_x f_y \cos \phi)$, where ϕ is the incidence angle. Report three quantities: $A_{\text{sensed}} = \sum a$ over sensed pixels; $A_{\text{noread}} = \sum a$ over no-reads inside the projected outline grown by 1 px; and $A_{\text{sensed}} + A_{\text{noread}}$, an upper bound for a reader that treats no-reads as part of the feature.
5. Sub-pixel area. For each pose, average its frames and interpolate h over valid pixels. Extract the $h = 0.5$ iso-contour with marching squares, deproject it onto the feature plane, and take the polygon area. Use this to check the per-frame pixel-count result for systematic error.
6. True area. $A_{\text{true}} = \pi D^2/4$, using the as-built D .
7. Geometric visible area. Ray-cast from the registered left camera, right camera, and projector centers through the as-built outline to the back plane. A_{geo} for a cutout is the overlap of the three projected outlines. Compute it with all three, and also with only the two cameras, because it is not known whether the VSX3000 needs projector light on a point to read it. Report which version tracks the data better. For a disk front face, $A_{\text{geo}} = A_{\text{true}}$.
8. Transfer curves. Plot $A_{\text{sensed}} / A_{\text{true}}$ and $A_{\text{sensed}} / A_{\text{geo}}$ against $D_{\text{px}} = D \cdot f_x / Z$, pooled over the features and stations of the Z sweep with the feature identity kept as a factor, for each gap

and feature type. Put disks and cutouts on the same axes so their asymmetry shows. Report the mean, the spread across poses (phase effect), and the spread within a pose (temporal effect).

9. Scaling test. Where two features overlap in D_{px} (half an octave at each junction of the ladder), compare their transfer curves over the shared range. Agreement within the bootstrap interval supports D_{px} as the governing variable. Disagreement is attributed to the Z-dependent noise $\sigma_{tot}(Z)$ and is reported as the difference between the two curves.
10. Edge bias. Convert to an equivalent diameter $D_s = 2\sqrt{(A_{sensed}/\pi)}$. The edge bias is $b = (D_s - D)/2$, in mm and in px. Fit b against Z using the largest feature, well above D_{50} , where b should not depend on D .
11. Consistency with B. Build a 2-D blur kernel from the H and V line spread functions (assumed separable). Blur an ideal disk of the as-built D with it and threshold at 0.5; this predicts the area. Agreement for large D and disagreement for small D means the sensor's interpolation fills in small features beyond what the edge spread function explains. Also check that the sign of b matches the edge offset s_{50} from 11.1.
12. Field and open-background variants. Compare the off-axis transfer curves with on-axis ones. For the open-background cutouts, compare the no-read area with A_{true} . Fill-in by front-plate values shows as $A_{noread} < A_{true}$.
13. Outputs. $C_{area_summary.csv}$, plus figures of the transfer curves, b vs Z , and phase-spread vs D_{px} .